

# Technical Information

## **AISWEI Interface**

(Based On Modbus Standard Protocol)

| Version | Comments                                  | Author         |
|---------|-------------------------------------------|----------------|
| 1.0     | First edition                             | Jincheng Wang  |
| 2.1     | Upgrade to 2.1 version                    | Jeff.Ji        |
| 2.1.1   | Upgrade to 2.1.1 version                  | Jeff.Ji        |
| 2.1.2   | Upgrade to 2.1.2 version                  | Guohao.Li      |
| 2.1.3   | Upgrade to 2.1.3 version                  | Shaochen.Mao   |
| 2.1.4   | Upgrade to 2.1.3 version: add the ct data | Zhengjian.zhou |

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‘Modbus’ herein refers to industrial standard serial communication protocol.

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# 1 Information on this Document

## Validity

This document is valid for AISWEI inverters.

## Target Group

This document is intended for qualified persons. Only persons with appropriate skills are allowed to perform the tasks described in this document.

## Terminology

| Information | Explanation                                                                     |
|-------------|---------------------------------------------------------------------------------|
| Pn          | The rated active power of device                                                |
| Pm          | The instantaneous power when the power control curve reaches the starting point |
| Sn          | The rated apparent power of device                                              |

## **2 Safety**

### **2.1 Intended Use**

The Modbus interface of the supported devices is designed for industrial use, via RS485 or RS422 protocol to enable remote control of the PV system, remote querying of values, and remote parameter setting.

### **2.2 Skills of Qualified Persons**

The activities described in this document must only be performed by qualified persons. Qualified persons must have the following skills:

- Detailed knowledge of the grid management services
- Knowledge of IP-based network protocols
- Training in the installation and configuration of IT systems
- Knowledge of the Modbus specifications
- Knowledge of and compliance with this document and all safety information

### 3 AISWEI Modbus Profile

#### 3.1 Information on the Assignment Tables

The assignment tables of the AISWEI Modbus profile present the following information:

| Information                    | Explanation                                                                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADR (DEC)                      | Decimal Modbus address, you need to remove 3x or 4x and subtract 1, then convert to hexadecimal and use it in the communication frame. Such as 31001 (decimal) → 1000 (decimal) → 0x03e8 (hexadecimal) |
| Description/<br>number code(s) | Short description of the Modbus registers and the number codes used.                                                                                                                                   |
| Type                           | Type of the data (see Section 3.2).                                                                                                                                                                    |
| Unit                           | Unit of the data.                                                                                                                                                                                      |
| Gain                           | Real value = Gain * output value                                                                                                                                                                       |
| Access                         | RO: Read Only<br>RW: Read and Write<br>WO: Write Only                                                                                                                                                  |

### 3.2 AISWEI Data Types and NaN Values

The following table shows the data types used in the AISWEI Modbus profile and the possible NaN values. The AISWEI data types are listed in the assignment tables in the Type column. They describe the data widths of the assigned values:

| Type   | Description                                                                                                                                                   | NaN Value   |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| B16    | Bit field (16-bit )                                                                                                                                           | 0xFFFF      |
| B32    | Bit field (32-bit )                                                                                                                                           | 0xFFFF FFFF |
| S16    | Signed integer (16-bit )                                                                                                                                      | 0x8000      |
| U16    | Unsigned integer (16-bit )                                                                                                                                    | 0xFFFF      |
| S32    | Signed integer (32-bit )                                                                                                                                      | 0x8000 0000 |
| U32    | Unsigned integer (32-bit )                                                                                                                                    | 0xFFFF FFFF |
| E16    | Number code ( 16-bit )                                                                                                                                        | 0xFFFF      |
| String | String type (16-bit, combination of two 8-bit ASCII characters, the high 8-bit is the first ASCII character, and the low 8-bit is the second ASCII character) | 0x0000      |



### 3.3 AISWEI Modbus Profile – Register Overview

In the following table you will find all the measured values and parameters of the AISWEI Modbus Profile.

#### Input Registers

| ADDR(DEC)   | Description/number code                                                                | Type   | Unit | Gain | Access |
|-------------|----------------------------------------------------------------------------------------|--------|------|------|--------|
| 31001       | Device Type:<br>1=Single phase / 3=Three pahse                                         | String | -    | -    | RO     |
| 31002       | Modbus address: Default as 3                                                           | U16    | -    | -    | RO     |
| 31003~31018 | Serial Number                                                                          | String | -    | -    | RO     |
| 31019~31026 | Machine type: for example, "ASW3000", please refer to the specific machine in practice | String | -    | -    | RO     |
| 31027       | Current grid code: refer to Section 3.5                                                | E16    | -    | -    | RO     |
| 31028~31029 | Rated Power                                                                            | U32    | W    | 1.0  | RO     |
| 31030~31036 | Master Software Version                                                                | String | -    | -    | RO     |
| 31037~31043 | Slave Software Version                                                                 | String | -    | -    | RO     |
| 31044~31050 | Safety Version                                                                         | String | -    | -    | RO     |
| 31057~31064 | Manufacturer's name: for example, "AISWEI", refer to the specific machine              | String | -    | -    | RO     |
| 31065~31072 | Brand name: for example, "AISWEI", please refer to the specific machine                | String | -    | -    | RO     |
| 31301       | Grid rated voltage                                                                     | U16    | V    | 0.1  | RO     |
| 31302       | Grid rated frequency                                                                   | U16    | Hz   | 0.01 | RO     |
| 31303~31304 | E-Today of inverter                                                                    | U32    | kWh  | 0.1  | RO     |
| 31305~31306 | E-Total of inverter                                                                    | U32    | kWh  | 0.1  | RO     |
| 31307~31308 | H-Total                                                                                | U32    | H    | 1.0  | RO     |
| 31309       | Device State:<br>0 = Wait<br>1 = Normal                                                | E16    | -    | -    | RO     |

|       |                                              |     |    |      |    |
|-------|----------------------------------------------|-----|----|------|----|
|       | 2 = Fault<br>4 = Checking                    |     |    |      |    |
| 31310 | Connect time                                 | U16 | s  | 1.0  | RO |
| 31311 | Air temperature                              | S16 | °C | 0.1  | RO |
| 31312 | Inverter U phase temperature                 | S16 | °C | 0.1  | RO |
| 31313 | Inverter V phase temperature                 | S16 | °C | 0.1  | RO |
| 31314 | Inverter W phase temperature                 | S16 | °C | 0.1  | RO |
| 31315 | Boost temperature                            | S16 | °C | 0.1  | RO |
| 31316 | Bidirectional DC/DC Converter temperature(*) | S16 | °C | 0.1  | RO |
| 31317 | Bus voltage                                  | U16 | V  | 0.1  | RO |
| 31319 | PV1 voltage                                  | U16 | V  | 0.1  | RO |
| 31320 | PV1 current                                  | U16 | A  | 0.01 | RO |
| 31321 | PV2 voltage                                  | U16 | V  | 0.1  | RO |
| 31322 | PV2 current                                  | U16 | A  | 0.01 | RO |
| 31323 | PV3 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31324 | PV3 current(*)                               | U16 | A  | 0.01 | RO |
| 31325 | PV4 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31326 | PV4 current(*)                               | U16 | A  | 0.01 | RO |
| 31327 | PV5 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31328 | PV5 current(*)                               | U16 | A  | 0.01 | RO |
| 31329 | PV6 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31330 | PV6 current(*)                               | U16 | A  | 0.01 | RO |
| 31331 | PV7 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31332 | PV7 current(*)                               | U16 | A  | 0.01 | RO |
| 31333 | PV8 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31334 | PV8 current(*)                               | U16 | A  | 0.01 | RO |
| 31335 | PV9 voltage(*)                               | U16 | V  | 0.1  | RO |
| 31336 | PV9 current(*)                               | U16 | A  | 0.01 | RO |
| 31337 | PV10 voltage(*)                              | U16 | V  | 0.1  | RO |
| 31338 | PV10 current(*)                              | U16 | A  | 0.01 | RO |
| 31339 | String 1 current(*)                          | U16 | A  | 0.1  | RO |
| 31340 | String 2 current(*)                          | U16 | A  | 0.1  | RO |

|             |                                 |     |     |      |    |
|-------------|---------------------------------|-----|-----|------|----|
| 31341       | String 3 current(*)             | U16 | A   | 0.1  | RO |
| 31342       | String 4 current(*)             | U16 | A   | 0.1  | RO |
| 31343       | String 5 current(*)             | U16 | A   | 0.1  | RO |
| 31344       | String 6 current(*)             | U16 | A   | 0.1  | RO |
| 31345       | String 7 current(*)             | U16 | A   | 0.1  | RO |
| 31346       | String 8 current(*)             | U16 | A   | 0.1  | RO |
| 31347       | String 9 current(*)             | U16 | A   | 0.1  | RO |
| 31348       | String 10 current(*)            | U16 | A   | 0.1  | RO |
| 31349       | String 11 current(*)            | U16 | A   | 0.1  | RO |
| 31350       | String 12 current(*)            | U16 | A   | 0.1  | RO |
| 31351       | String 13 current(*)            | U16 | A   | 0.1  | RO |
| 31352       | String 14 current(*)            | U16 | A   | 0.1  | RO |
| 31353       | String 15 current(*)            | U16 | A   | 0.1  | RO |
| 31354       | String 16 current(*)            | U16 | A   | 0.1  | RO |
| 31355       | String 17 current(*)            | U16 | A   | 0.1  | RO |
| 31356       | String 18 current(*)            | U16 | A   | 0.1  | RO |
| 31357       | String 19 current(*)            | U16 | A   | 0.1  | RO |
| 31358       | String 20 current(*)            | U16 | A   | 0.1  | RO |
| 31359       | L1 Phase voltage                | U16 | V   | 0.1  | RO |
| 31360       | L1 Phase current                | U16 | A   | 0.1  | RO |
| 31361       | L2 Phase voltage(*)             | U16 | V   | 0.1  | RO |
| 31362       | L2 Phase current(*)             | U16 | A   | 0.1  | RO |
| 31363       | L3 Phase voltage(*)             | U16 | V   | 0.1  | RO |
| 31364       | L3 Phase current(*)             | U16 | A   | 0.1  | RO |
| 31365       | RS Line voltage(*)              | U16 | V   | 0.1  | RO |
| 31366       | RT Line voltage(*)              | U16 | V   | 0.1  | RO |
| 31367       | ST Line voltage(*)              | U16 | V   | 0.1  | RO |
| 31368       | Grid frequency                  | U16 | Hz  | 0.01 | RO |
| 31369~31370 | Apparent power                  | U32 | VA  | 1.0  | RO |
| 31371~31372 | Active power                    | S32 | W   | 1.0  | RO |
| 31373~31374 | Reactive power                  | S32 | Var | 1.0  | RO |
| 31375       | Power factor                    | S16 | -   | 0.01 | RO |
| 31377       | Fault state of inverter itself: | E16 | -   | -    | RO |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |     |     |    |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----|
|             | 0 = No internal fault<br>1 = Internal fault                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |     |     |    |
| 31378       | Error message:please refer to section 3.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E16 | -   | -   | RO |
| 31379       | Warning message:please refer to section 3.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | E16 | -   | -   | RO |
| 31601~31602 | PV total power                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | U32 | W   | -   | RO |
| 31603~31604 | PV E-Today                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | U32 | kWh | 0.1 | RO |
| 31605~31606 | PV E-Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | U32 | kWh | 0.1 | RO |
| 31607       | Battery communication status:<br>0x000A=Normal<br>0x0005=Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | E16 | -   | -   | RO |
| 31608       | Battery status:<br>0 = Not available<br>1 = Idle<br>2 = Charging<br>3 = Discharging<br>4 = Error                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | E16 | -   | -   | RO |
| 31609       | Battery Error Status 1:<br>bit0 communication data error 0-valid 1-invalid<br>bit1 cell or module voltage is too high 0-valid 1-invalid<br>bit2 Cell or module voltage is too low 0-valid 1-invalid<br>bit3 battery temperature is too high 0-valid 1-invalid<br>bit4 battery temperature is too low 0-valid 1-invalid<br>bit5 discharging current over limit 0-valid 1-invalid<br>bit6 charging current over limit 0-valid 1-invalid<br>bit7 internal communication error 0-valid 1-invalid<br>bit8 Internal cell voltage is unbalance 0-valid 1-invalid<br>bit9 System insulation resistance is too low 0-valid 1-invalid<br>bit10 voltage sensor failure 0-valid 1-invalid | B16 | -   | -   | RO |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |   |   |    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|----|
|       | bit11 temperature sensor failure 0-valid 1-invalid<br>bit12 contactor failure 0-valid 1-invalid<br>bit13 Self-test failure during starting 0-valid 1-invalid<br>bit14 IC self-test failure 0-valid 1-invalid<br><br>The default value of undefined bit is 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |   |   |    |
| 31610 | Battery Error Status 2:<br>bit0 Self-test failure for battery voltage 0-valid 1-failure<br>bit1 Self-test failure for system voltage 0-valid 1-invalid<br>bit2 Self-test failure for system insulation resistance 0-valid 1-invalid<br>bit3 RTC invalid 0-valid 1-invalid<br>bit4 EEPROM failure 0-valid 1-invalid<br>bit5 Flash failure 0-valid 1-invalid<br>bit6 AFE invalid 0-valid 1-invalid<br>bit7 Chip failure for insulation resistance detect 0-valid 1-invalid<br>bit9 Chip failure for current sampling 0-valid 1-invalid<br>bit10 HDC failure 0-valid 1-invalid<br>bit11 Daisy chain failure 0-valid 1-invalid<br>bit12 Failure for precharge 0-valid 1-invalid<br><br>The default value of undefined bit is 1 | B16 | - | - | RO |
| 31613 | Battery warning status 1:<br>bit0 Communication data error 0-valid 1-invalid<br>bit1 Battery or module voltage is too high 0-valid 1-invalid<br>bit2 Battery or module voltage is too low 0-valid 1-invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | B16 | - | - | RO |

|             |                                                                                                                                                                                                                                                                                                                                                                                                     |     |     |      |    |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|----|
|             | bit3 Battery temperature is too high 0-valid 1-invalid<br>bit4 Battery temperature is too low 0-valid 1-invalid<br>bit5 discharging current over limit 0-valid 1-invalid<br>bit6 charging current over limit 0-valid 1-invalid<br>bit7 Internal communication failure 0-valid 1-invalid<br>bit8 Internal cell voltage is unbalance 0-valid 1-invalid<br><br>The default value of undefined bit is 1 |     |     |      |    |
| 31617       | Battery voltage                                                                                                                                                                                                                                                                                                                                                                                     | U16 | V   | 0.01 | RO |
| 31618       | Battery current                                                                                                                                                                                                                                                                                                                                                                                     | S16 | A   | 0.1  | RO |
| 31619~31620 | Battery power                                                                                                                                                                                                                                                                                                                                                                                       | S32 | w   | 1    | RO |
| 31621       | Battery temperature                                                                                                                                                                                                                                                                                                                                                                                 | S16 | °C  | 0.1  | RO |
| 31622       | Battery SOC                                                                                                                                                                                                                                                                                                                                                                                         | U16 | -   | 0.01 | RO |
| 31623       | Battery SOH                                                                                                                                                                                                                                                                                                                                                                                         | U16 | -   | 0.01 | RO |
| 31624       | Battery charging current limit                                                                                                                                                                                                                                                                                                                                                                      | U16 | A   | 0.1  | RO |
| 31625       | Battery discharge current limit                                                                                                                                                                                                                                                                                                                                                                     | U16 | A   | 0.1  | RO |
| 31626~31627 | Battery E-Charge-Today                                                                                                                                                                                                                                                                                                                                                                              | U32 | kWh | 0.1  | RO |
| 31628~31629 | Battery E-Discharge-Today                                                                                                                                                                                                                                                                                                                                                                           | U32 | kWh | 0.1  | RO |
| 31630~31631 | E-Consumption-Today at AC side                                                                                                                                                                                                                                                                                                                                                                      | U32 | kWh | 0.1  | RO |
| 31632~31633 | E-Generation-Today at AC side                                                                                                                                                                                                                                                                                                                                                                       | U32 | kWh | 0.1  | RO |
| 31634       | EPS load voltage                                                                                                                                                                                                                                                                                                                                                                                    | U16 | V   | 0.1  | RO |
| 31635       | EPS load current                                                                                                                                                                                                                                                                                                                                                                                    | U16 | A   | 0.1  | RO |
| 31636       | EPS load frequency                                                                                                                                                                                                                                                                                                                                                                                  | U16 | Hz  | 0.01 | RO |
| 31637~31638 | EPS load active power                                                                                                                                                                                                                                                                                                                                                                               | U32 | w   | 1    | RO |
| 31639~31640 | EPS load reactive power                                                                                                                                                                                                                                                                                                                                                                             | U32 | Var | 1    | RO |
| 31641~31642 | E-Consumption-Today at EPS load side                                                                                                                                                                                                                                                                                                                                                                | U32 | kWh | 0.1  | RO |
| 31643~31644 | E-Consumption-Total at EPS load side                                                                                                                                                                                                                                                                                                                                                                | U32 | kWh | 0.1  | RO |
| 31645       | Phase 1 voltage for EPS Load                                                                                                                                                                                                                                                                                                                                                                        | U16 | V   | 0.1  | RO |

|             |                                     |     |     |     |    |
|-------------|-------------------------------------|-----|-----|-----|----|
| 31646       | Phase 1 current for EPS Load        | U16 | A   | 0.1 | RO |
| 31647       | Phase 2 voltage for EPS Load        | U16 | V   | 0.1 | RO |
| 31648       | Phase 2 current for EPS Load        | U16 | A   | 0.1 | RO |
| 31649       | Phase 3 voltage for EPS Load        | U16 | V   | 0.1 | RO |
| 31650       | Phase 3 current for EPS Load        | U16 | A   | 0.1 | RO |
| 31651~31652 | Phase 1 active power for EPS Load   | U32 | w   | 1   | RO |
| 31653~31654 | Phase 1 reactive power for EPS Load | S32 | Var | 1   | RO |
| 31655~31656 | Phase 2 active power for EPS Load   | U32 | w   | 1   | RO |
| 31657~31658 | Phase 2 reactive power for EPS Load | S32 | Var | 1   | RO |
| 31659~31660 | Phase 3 active power for EPS Load   | U32 | w   | 1   | RO |
| 31661~31662 | Phase 3 reactive power for EPS Load | S32 | Var | 1   | RO |
| 31663~31664 | Phase 1 active power for Grid       | U32 | w   | 1   | RO |
| 31665~31666 | Phase 1 reactive power for Grid     | S32 | Var | 1   | RO |
| 31667~31668 | Phase 2 active power for Grid       | U32 | w   | 1   | RO |
| 31669~31670 | Phase 2 reactive power for Grid     | S32 | Var | 1   | RO |
| 31671~31672 | Phase 3 active power for Grid       | U32 | w   | 1   | RO |
| 31673~31674 | Phase 3 reactive power for Grid     | S32 | Var | 1   | RO |
| 31675~31676 | Energy charge today for Grid        | U32 | kWh | 0.1 | RO |
| 31677~31678 | Energy charge total for Grid        | U32 | kWh | 0.1 | RO |
| 31679       | Battery insulation resistance       | U16 | kΩ  | 1   | RO |
| 31680       | Battery charge/discharge cycles     | U16 | -   | 1   | RO |
| 31681       | Environment temperature             | U16 | %   | 0.1 | RO |

**Holding register**

| ADDR(DEC) | Description/number code                                                                                                                                                                                   | Type | Unit | Gain | Access |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|--------|
| 40201     | Remote switch command:<br>0 = POWER OFF<br>1 = POWER ON<br>170 = Initialization status                                                                                                                    | E16  | -    | -    | RW     |
| 41001     | RTC:Year                                                                                                                                                                                                  | U16  | -    | -    | RW     |
| 41002     | RTC:Month                                                                                                                                                                                                 | U16  | -    | -    | RW     |
| 41003     | RTC:Day                                                                                                                                                                                                   | U16  | -    | -    | RW     |
| 41004     | RTC:Hour                                                                                                                                                                                                  | U16  | -    | -    | RW     |
| 41005     | RTC:Minute                                                                                                                                                                                                | U16  | -    | -    | RW     |
| 41006     | RTC:Seconds                                                                                                                                                                                               | U16  | -    | -    | RW     |
| 41102     | Storage Inverter Switch:<br>1 - OFF<br>2 - ON                                                                                                                                                             | E16  | -    | -    | RW     |
| 41103     | Type selection of energy storage machine:<br>0 - Invalid<br>1 - Energy storage machine<br>2 - Grid off inveter<br>3 - Grid connected inverter<br>4 - Force charge with City electricity (battery wake-up) | E16  | -    | -    | RW     |
| 41104     | Run mode:<br>0 - Invalid<br>1 - Off<br>2 - Self generating self use<br>3 - Backup power supply<br>4 - Customer defined                                                                                    | E16  | -    | -    | RW     |
| 41105     | Battery manufacturer:                                                                                                                                                                                     | E16  | -    | -    | RW     |



|             |                                                                                                                                     |     |   |   |    |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------|-----|---|---|----|
|             | 1 - PYLON<br>2 - DYNESS<br>3 - BYD<br>4 - LG<br>5 - AISWEI                                                                          |     |   |   |    |
| 41108       | Smart meter status:<br>0x000A - Meter Online<br>0x0005 - Meter Offline                                                              | E16 | - | - | RW |
| 41109       | Smart meter adjustment<br>flag bit:<br>0x000A = Start<br>0x0005 = Stop                                                              | E16 | - | - | RW |
| 41110~41111 | Set target power value                                                                                                              | S32 | w | 1 | RW |
| 41112~41113 | Current power value of<br>smart meter                                                                                               | S32 | w | 1 | RW |
| 41114       | Anti reverse current<br>flag:<br>0x000A = ON<br>0x0005 = OFF                                                                        | E16 | - | - | RW |
| 41115       | Battery wake-up (Force<br>charge) sign:<br>0x000A = ON<br>0x0005 = OFF<br>0xFFFF =Not triggered                                     | E16 | - | - | RW |
| 41116       | UPS function:<br>0 = Enable EPS function<br>1 = Enable UPS function                                                                 | U16 | - | - | RW |
| 41151       | Commbox and cloud<br>communication status:<br>0x000A = Cloud Online<br>0x0005 = Cloud Offline<br>0x00AF = Network not<br>configured | E16 | - | - | RW |
| 41152       | Charge discharge flag<br>bit:<br>1 - Stop<br>2 - Charging<br>3 - Discharge                                                          | E16 | - | - | RW |
| 41153       | Charge and discharge<br>power command:<br>' - ' - charging power                                                                    | S16 | W | 1 | RW |

|       |                                                                              |     |   |      |    |
|-------|------------------------------------------------------------------------------|-----|---|------|----|
|       | ' + ' - discharge power                                                      |     |   |      |    |
| 41154 | Charging SOC upper limit                                                     | U16 | % | 0.01 | RW |
| 41155 | Discharge SOC lower limit                                                    | U16 | % | 0.01 | RW |
| 41156 | Obtaining power ratio of power grid                                          | U16 | % | 0.01 | RW |
| 44001 | Active power control function:<br>0 = Disable<br>1 = Enable                  | E16 | - | -    | RW |
| 44002 | EEG control function:<br>0 = Disable<br>1 = Enable                           | E16 | - | -    | RW |
| 44003 | Slope load function:<br>0 = Disable<br>1 = Enable                            | E16 | - | -    | RW |
| 44004 | Overvoltage reduce power function:<br>0 = Disable<br>1 = Enable              | E16 | - | -    | RW |
| 44005 | Overfrequency reduce power function:<br>0 = Disable<br>1 = Enable            | E16 | - | -    | RW |
| 44006 | Reactive power control function:<br>0 = Disable<br>1 = Enable                | E16 | - | -    | RW |
| 44007 | LVRT Function:<br>0 = Disable<br>1 = Enable                                  | E16 | - | -    | RW |
| 44008 | HVRT Function:<br>0 = Disable<br>1 = Enable                                  | E16 | - | -    | RW |
| 44009 | 10 Minutes Average Overvoltage protect function<br>0 = Disable<br>1 = Enable | E16 | - | -    | RW |
| 44010 | Islanding protect function:                                                  | E16 | - | -    | RW |

|       |                                                                        |     |   |   |    |
|-------|------------------------------------------------------------------------|-----|---|---|----|
|       | 0 = Disable<br>1 = Enable                                              |     |   |   |    |
| 44012 | PE connection check function:<br>0 = Disable<br>1 = Enable             | E16 | - | - | RW |
| 44014 | AFCI function:<br>0 = Disable<br>1 = Enable                            | E16 | - | - | RW |
| 44015 | PV string current monitoring function:<br>0 = Disable<br>1 = Enable    | E16 | - | - | RW |
| 44017 | Overload function:<br>0 = Disable<br>1 = Enable                        | E16 | - | - | RW |
| 44019 | SPD detection function:<br>0 = Disable<br>1 = Enable                   | E16 | - | - | RW |
| 44020 | Low voltage increase power function:<br>0 = Disable<br>1 = Enable      | E16 | - | - | RW |
| 44021 | Low frequency increase power function:<br>0 = Disable<br>1 = Enable    | E16 | - | - | RW |
| 44023 | Primary low frequency function(*):<br>0 = Disable<br>1 = Enable        | E16 | - | - | RW |
| 44024 | Communication loss detection function(*):<br>0 = Disable<br>1 = Enable | E16 | - | - | RW |
| 44025 | Shadow MPPT function:<br>0 = Disable<br>1 = Enable                     | E16 | - | - | RW |
| 44026 | External input signal function:                                        | E16 | - | - | RW |

|             |                                                                   |     |    |      |    |
|-------------|-------------------------------------------------------------------|-----|----|------|----|
|             | 0 = Disable<br>1 = Enable                                         |     |    |      |    |
| 44027       | Sunspec write function:<br>0 = Disable<br>1 = Enable              | E16 | -  | -    | RW |
| 45201       | Grid code: please refer<br>to section 3.5                         | E16 | -  | -    | RW |
| 45202       | Overvoltage protection<br>value of the first grid con-<br>nection | U16 | V  | 0.1  | RW |
| 45203       | Overvoltage protection<br>value of the first grid con-<br>nection | U16 | V  | 0.1  | RW |
| 45204       | Overvoltage protection<br>value of the first grid con-<br>nection | U16 | Hz | 0.01 | RW |
| 45205       | Underfrequency protec-<br>tion value for first grid<br>connection | U16 | Hz | 0.01 | RW |
| 45206       | Grid Voltage High Limit3                                          | U16 | V  | 0.1  | RW |
| 45207~45208 | Grid Voltage High Limit<br>Time3                                  | U32 | ms | 1.0  | RW |
| 45209       | Grid Voltage High Limit2                                          | U16 | V  | 0.1  | RW |
| 45210~45211 | Grid Voltage High Limit<br>Time2                                  | U32 | ms | 1.0  | RW |
| 45212       | Grid Voltage High Limit1                                          | U16 | V  | 0.1  | RW |
| 45213~45214 | Grid Voltage High Limit<br>Time1                                  | U32 | ms | 1.0  | RW |
| 45215       | Grid Voltage Low Limit3                                           | U16 | V  | 0.1  | RW |
| 45216~45217 | Grid Voltage Low Limit<br>Time3                                   | U32 | ms | 1.0  | RW |
| 45218       | Grid Voltage Low Limit2                                           | U16 | V  | 0.1  | RW |
| 45219~45220 | Grid Voltage Low Limit<br>Time2                                   | U32 | ms | 1.0  | RW |
| 45221       | Grid Voltage Low Limit1                                           | U16 | V  | 0.1  | RW |
| 45222~45223 | Grid Voltage Low Limit<br>Time1                                   | U32 | ms | 1.0  | RW |

|             |                                             |     |      |      |    |
|-------------|---------------------------------------------|-----|------|------|----|
| 45224       | 10 Minutes Average Overvoltage Threshold    | U16 | V    | 0.1  | RW |
| 45225       | 10 Minutes Average Overvoltage Portect Time | U16 | ms   | 1.0  | RW |
| 45226       | Overvoltage recover value                   | U16 | V    | 0.1  | RW |
| 45227       | Undervoltage recover value                  | U16 | V    | 0.1  | RW |
| 45228       | Grid Frequency High Limit3                  | U16 | Hz   | 0.01 | RW |
| 45229~45230 | Grid Frequency High Limit Time3             | U32 | ms   | 1.0  | RW |
| 45231       | Grid Frequency High Limit2                  | U16 | Hz   | 0.01 | RW |
| 45232~45233 | Grid Frequency High Limit Time2             | U32 | ms   | 1.0  | RW |
| 45234       | Grid Frequency High Limit1                  | U16 | Hz   | 0.01 | RW |
| 45235~45236 | Grid Frequency High Limit Time1             | U32 | ms   | 1.0  | RW |
| 45237       | Grid Frequency Low Limit3                   | U16 | Hz   | 0.01 | RW |
| 45238~45239 | Grid Frequency Low Limit Time3              | U32 | ms   | 1.0  | RW |
| 45240       | Grid Frequency Low Limit2                   | U16 | Hz   | 0.01 | RW |
| 45241~45242 | Grid Frequency Low Limit Time2              | U32 | ms   | 1.0  | RW |
| 45243       | Grid Frequency Low Limit1                   | U16 | Hz   | 0.01 | RW |
| 45244~45245 | Grid Frequency Low Limit Time1              | U32 | ms   | 1.0  | RW |
| 45246       | Vary rate of Frequecny protect value        | U16 | Hz/s | 0.01 | RW |
| 45247~45248 | Vary rate of Frequecny protect time         | U32 | ms   | 1.0  | RW |
| 45249       | Overfrequency recover value                 | U16 | Hz   | 0.01 | RW |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |         |      |    |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|------|----|
| 45250 | Underfrequency recover value                                                                                                                                                                                                                                                                                                                                                                                                                  | U16 | Hz      | 0.01 | RW |
| 45251 | Time of first connection to grid                                                                                                                                                                                                                                                                                                                                                                                                              | U16 | s       | 1.0  | RW |
| 45252 | Time of re-connection to grid                                                                                                                                                                                                                                                                                                                                                                                                                 | U16 | s       | 1.0  | RW |
| 45253 | ISO protect threshold                                                                                                                                                                                                                                                                                                                                                                                                                         | U16 | kΩ      | 1.0  | RW |
| 45254 | DCI protect threshold                                                                                                                                                                                                                                                                                                                                                                                                                         | U16 | mA      | 1.0  | RW |
| 45255 | DCI protect time                                                                                                                                                                                                                                                                                                                                                                                                                              | U16 | ms      | 1.0  | RW |
| 45401 | Load rate of first connection to grid                                                                                                                                                                                                                                                                                                                                                                                                         | U16 | %Pn/min | 1.0  | RW |
| 45402 | Load rate of re-connection to grid                                                                                                                                                                                                                                                                                                                                                                                                            | U16 | %Pn/min | 1.0  | RW |
| 45403 | Active Power Set                                                                                                                                                                                                                                                                                                                                                                                                                              | U16 | %Pn     | 0.01 | RW |
| 45404 | Increase rate of active power                                                                                                                                                                                                                                                                                                                                                                                                                 | U16 | %Pn/min | 0.01 | RW |
| 45405 | Decrease rate of active power                                                                                                                                                                                                                                                                                                                                                                                                                 | U16 | %Pn/min | 0.01 | RW |
| 45408 | Over frequency reduce power mode:<br>0 = None<br>1 = Fixed reduction ratio, non – hysteresis<br>2 = Fixed reduction ratio, hysteresis<br>3 = Not fixed reduction ratio, non – hysteresis<br>4 = Not fixed reduction ratio, hysteresis<br>5 = Three points over frequency reduce power, non – hysteresis<br>6 = Three points over frequency reduce power, hysteresis<br>7 = Energy storage Italy over frequency reduce power, non – hysteresis | E16 | -       | -    | RW |

|       |                                                                                                                                                                                                                                                                                                                                              |     |          |      |    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|------|----|
|       | 8 = Energy storage Italy<br>over frequency reduce<br>power, hysteresis                                                                                                                                                                                                                                                                       |     |          |      |    |
| 45409 | Over frequency reduce<br>power:<br>Start frequency                                                                                                                                                                                                                                                                                           | U16 | Hz       | 0.01 | RW |
| 45410 | Over frequency reduce<br>power:<br>Stop frequency                                                                                                                                                                                                                                                                                            | U16 | Hz       | 0.01 | RW |
| 45411 | Over frequency reduce<br>power:<br>Back frequency                                                                                                                                                                                                                                                                                            | U16 | Hz       | 0.01 | RW |
| 45412 | The reduce ratio of over<br>frequency reduce power                                                                                                                                                                                                                                                                                           | U16 | %Pnor%Pm | 0.01 | RW |
| 45413 | Over frequency reduce<br>power :reduce power de-<br>lay time                                                                                                                                                                                                                                                                                 | U16 | s        | 0.1  | RW |
| 45414 | Over frequency reduce<br>power:recover power de-<br>lay time                                                                                                                                                                                                                                                                                 | U16 | s        | 0.1  | RW |
| 45416 | Speed of Over frequency<br>recover to Pn                                                                                                                                                                                                                                                                                                     | U16 | %Pn/min  | 0.01 | RW |
| 45417 | Over frequency reduce<br>power(*):<br>0 power frequency point                                                                                                                                                                                                                                                                                | U16 | Hz       | 0.01 | RW |
| 45419 | Over voltage reduce<br>power mode:<br>0 = None<br>1 = Not fixed reduction<br>ratio, non – hysteresis<br>2 = Not fixed reduction<br>ratio, hysteresis<br>3 = Fixed reduction ratio,<br>non – hysteresis<br>4 = Fixed reduction ratio,<br>hysteresis<br>5 = Taiwan's autonomous<br>power regulation<br>6 = Trina Solar customiza-<br>tion mode | E16 | -        | -    | RW |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |          |      |    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|------|----|
| 45420 | Over voltage reduce power:<br>Start voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | U16 | %Un      | 0.01 | RW |
| 45422 | Over voltage reduce power:<br>Stop voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | U16 | %Un      | 0.01 | RW |
| 45424 | Over voltage reduce power:<br>Back voltage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | U16 | %Un      | 0.01 | RW |
| 45426 | The reduce ratio of over voltage reduce power                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | U16 | %Pnor%Pm | 0.01 | RW |
| 45427 | Over voltage reduce power delay time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | U16 | s        | 0.1  | RW |
| 45428 | Over voltage recover power delay time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | U16 | s        | 0.1  | RW |
| 45429 | Speed of Over voltage re-cover to Pn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | U16 | %Pn/min  | 0.01 | RW |
| 45432 | Under frequency increase power mode:<br>0 = None<br>1 = Fixed reduction ratio, non – hysteresis<br>2 = Fixed reduction ratio, hysteresis<br>3 = Not fixed reduction ratio, non – hysteresis<br>4 = Not fixed reduction ratio, hysteresis<br>5 = Three points under frequency increase power, non – hysteresis<br>6 = Three points under frequency increase power, hysteresis<br>7 = Energy storage Italy under frequency increase power, non – hysteresis<br>8 = Energy storage Italy under frequency increase power, hysteresis | E16 | -        | -    | RW |



|       |                                                                                                                                                                                                                                    |     |          |      |    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|------|----|
| 45433 | Under frequency increase power:<br>Start frequency                                                                                                                                                                                 | U16 | Hz       | 0.01 | RW |
| 45434 | Under frequency increase power:<br>Stop frequency                                                                                                                                                                                  | U16 | Hz       | 0.01 | RW |
| 45435 | Under frequency increase power:<br>Back frequency                                                                                                                                                                                  | U16 | Hz       | 0.01 | RW |
| 45436 | The increase ratio of under frequency increase power                                                                                                                                                                               | U16 | %Pnor%Pm | 0.01 | RW |
| 45437 | Under frequency increase power delay time                                                                                                                                                                                          | U16 | s        | 0.1  | RW |
| 45438 | Under frequency recover power delay time                                                                                                                                                                                           | U16 | s        | 0.1  | RW |
| 45440 | Speed of Under frequency recover to Pn                                                                                                                                                                                             | U16 | %Pn/min  | 0.01 | RW |
| 45441 | Under frequency increase power 0 power frequency point                                                                                                                                                                             | U16 | Hz       | 0.01 | RW |
| 45443 | Under voltage increase power mode:<br>0 = None<br>1 = Fixed increase ratio, non – hysteresis<br>2 = Fixed increase ratio, hysteresis<br>3 = Not fixed increase ratio, non – hysteresis<br>4 = Not fixed increase ratio, hysteresis | E16 | -        | -    | RW |
| 45444 | Under voltage increase power:<br>Start voltage                                                                                                                                                                                     | U16 | %Un      | 0.01 | RW |
| 45445 | Under voltage increase power:<br>Stop voltage                                                                                                                                                                                      | U16 | %Un      | 0.01 | RW |
| 45446 | Under voltage increase power:                                                                                                                                                                                                      | U16 | %Un      | 0.01 | RW |

|       |                                                                                                                                                                                                                                                                |     |          |        |    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|--------|----|
|       | Back voltage                                                                                                                                                                                                                                                   |     |          |        |    |
| 45447 | The increase ratio of under voltage increase power                                                                                                                                                                                                             | U16 | %Pnor%Pm | 0.01   | RW |
| 45448 | Under voltage increase power delay time                                                                                                                                                                                                                        | U16 | s        | 0.1    | RW |
| 45449 | Under voltage increase power delay time                                                                                                                                                                                                                        | U16 | s        | 0.1    | RW |
| 45450 | Speed of under voltage recover to Pn                                                                                                                                                                                                                           | U16 | %Pn/min  | 0.01   | RW |
| 45451 | Pav(*)                                                                                                                                                                                                                                                         | S16 | %Pn      | 0.01   | RW |
| 45452 | DRMs Pval(*)                                                                                                                                                                                                                                                   | U16 | %Pn      | 0.01   | RW |
| 45501 | Reactive power control mode:<br>0 = None<br>1 = Fixed power factor<br>2 = $\cos \varphi(P)$ curve<br>3 = Fixed Q value<br>4 = Fixed Q value of AU DRMs<br>5 = Linear Q(U) curve<br>6 = Hysteresis Q(U) curve<br>7 = Taiwan's autonomous control and regulation | E16 | -        | -      | RW |
| 45502 | Time constant of reactive power curve                                                                                                                                                                                                                          | U16 | s        | 1.0    | RW |
| 45503 | Power factor                                                                                                                                                                                                                                                   | S16 | -        | 0.0001 | RW |
| 45504 | $\cos \varphi(P)$ curve:<br>Active power of the first point                                                                                                                                                                                                    | U16 | %Pn      | 0.01   | RW |
| 45505 | $\cos \varphi(P)$ curve:<br>$\cos \varphi$ of the first point                                                                                                                                                                                                  | S16 | -        | 0.0001 | RW |
| 45506 | $\cos \varphi(P)$ curve:<br>Active power of the second point                                                                                                                                                                                                   | U16 | %Pn      | 0.01   | RW |
| 45507 | $\cos \varphi(P)$ curve:<br>$\cos \varphi$ of the second point                                                                                                                                                                                                 | S16 | -        | 0.0001 | RW |
| 45508 | $\cos \varphi(P)$ curve:                                                                                                                                                                                                                                       | U16 | %Pn      | 0.01   | RW |

|       |                                                                       |     |     |        |    |
|-------|-----------------------------------------------------------------------|-----|-----|--------|----|
|       | Active power of the third point                                       |     |     |        |    |
| 45509 | cos $\varphi$ (P) curve:<br>cos $\varphi$ of the third point          | S16 | -   | 0.0001 | RW |
| 45510 | cos $\varphi$ (P) curve:<br>Active power of the fourth point          | U16 | %Pn | 0.01   | RW |
| 45511 | cos $\varphi$ (P) curve:<br>cos $\varphi$ of the fourth point         | S16 | -   | 0.0001 | RW |
| 45512 | Lock in voltage (for cos $\varphi$ (P) curve)                         | U16 | %Un | 0.01   | RW |
| 45513 | Lock out voltage (for cos $\varphi$ (P) curve)                        | U16 | %Un | 0.01   | RW |
| 45516 | Q Set Value                                                           | S16 | %Sn | 0.01   | RW |
| 45518 | Q(U) curve:<br>U of the first point                                   | U16 | %Un | 0.01   | RW |
| 45519 | Q(U) curve:<br>Q of the first point                                   | S16 | %Sn | 0.01   | RW |
| 45520 | Q(U) curve:<br>U of the second point                                  | U16 | %Un | 0.01   | RW |
| 45521 | Q(U) curve:<br>Q of the second point                                  | S16 | %Sn | 0.01   | RW |
| 45522 | Q(U) curve:<br>U of the third point                                   | U16 | %Un | 0.01   | RW |
| 45523 | Q(U) curve:<br>Q of the third point                                   | S16 | %Sn | 0.01   | RW |
| 45524 | Q(U) curve:<br>U of the fourth point                                  | U16 | %Un | 0.01   | RW |
| 45525 | Q(U) curve:<br>Q of the fourth point                                  | S16 | %Sn | 0.01   | RW |
| 45526 | Lock in power (for Q(U) curve)                                        | U16 | %Pn | 0.01   | RW |
| 45527 | Lock outpower (for Q(U) curve)                                        | U16 | %Pn | 0.01   | RW |
| 45601 | LVRT reactive current calculation mode:<br>0 = None<br>1 = GB/T 19964 | E16 | -   | -      | RW |

|       |                                                                                                                                   |     |     |      |    |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|----|
|       | 2 = BDEW                                                                                                                          |     |     |      |    |
| 45602 | LVRT three phase fault reactive current limit                                                                                     | U16 | %In | 0.01 | RW |
| 45603 | LVRT single/double phase fault reactive current limit                                                                             | U16 | %In | 0.01 | RW |
| 45604 | LVRT fault detection voltage type:<br>0 = Phase voltage<br>1 = Line voltage<br>2 = Positive sequence voltage                      | E16 | -   | -    | RW |
| 45605 | Threshold of insensitive area of positive sequence voltage jump                                                                   | U16 | %Un | 0.1  | RW |
| 45606 | LVRT Trigger voltage                                                                                                              | U16 | %Un | 0.01 | RW |
| 45607 | K-factor of positive sequence reactive current                                                                                    | U16 | -   | 0.01 | RW |
| 45608 | reactive power maintenance after LVRT voltage recovery                                                                            | U16 | ms  | 1    | RW |
| 45609 | LVRT active power limit mode:<br>0 = Active power first<br>1 = Active power reduced to below 10% In<br>2 = Unlimited active power | E16 | -   | -    | RW |
| 45610 | HVRT reactive current calculation mode:<br>0 = None<br>1 = GB/T 19964<br>2 = BDEW                                                 | E16 | -   | -    | RW |
| 45611 | HVRT three phase fault reactive current limit                                                                                     | U16 | %In | 0.01 | RW |
| 45612 | HVRT single/double phase fault reactive current limit                                                                             | U16 | %In | 0.01 | RW |
| 45613 | HVRT fault detection voltage type:                                                                                                | E16 | -   | -    | RW |

|       |                                                                                                                                   |     |     |      |    |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|----|
|       | 0 = Phase voltage<br>1 = Line voltage<br>2 = Positive sequence voltage                                                            |     |     |      |    |
| 45614 | Threshold of insensitive area of negative sequence voltage jump                                                                   | U16 | %Un | 0.1  | RW |
| 45615 | HVRT Trigger voltage                                                                                                              | U16 | %Un | 0.01 | RW |
| 45616 | K-factor of negative sequence reactive current                                                                                    | U16 | -   | 0.01 | RW |
| 45617 | reactive power maintenance after HVRT voltage recovery                                                                            | U16 | ms  | 1    | RW |
| 45618 | HVRT active power limit mode:<br>0 = Active power first<br>1 = Active power reduced to below 10% In<br>2 = Unlimited active power | E16 | -   | -    | RW |
| 45619 | Zero current threshold                                                                                                            | U16 | %Un | 0.01 | RW |
| 46520 | AFCI self-test status:<br>0 = self-test failed<br>1 = self-test successful                                                        | E16 | -   | -    | RO |
| 46521 | AFCI detection sensitivity settings:<br>0 = high detection accuracy<br>1 = low detection accuracy                                 | E16 | -   | -    | RW |
| 46522 | AFCI fault reset method selection:<br>0 = automatic mode<br>1 = manual mode                                                       | E16 | -   | -    | RW |
| 46523 | Reset faults manually:<br>0 = no need<br>1 = Manual clear fault (active in arc fault and manual reset mode)                       | E16 | -   | -    | RW |

## CT Data

|       |                               |     |     |      |    |
|-------|-------------------------------|-----|-----|------|----|
| 46401 | Phase 1 line to neutral volts | U16 | V   | 0.1  | RW |
| 46402 | Phase 2 line to neutral volts | U16 | V   | 0.1  | RW |
| 46403 | Phase 3 line to neutral volts | U16 | V   | 0.1  | RW |
| 46404 | Phase 1 current               | U16 | A   | 0.1  | RW |
| 46405 | Phase 2 current               | U16 | A   | 0.1  | RW |
| 46406 | Phase 3 current               | U16 | A   | 0.1  | RW |
| 46407 | Phase 1 power                 | S32 | w   | 1    | RW |
| 46409 | Phase 2 power                 | S32 | w   | 1    | RW |
| 46411 | Phase 3 power                 | S32 | w   | 1    | RW |
| 46413 | Phase 1 volt amps             | U32 | VA  | 1    | RW |
| 46415 | Phase 2 volt amps             | U32 | VA  | 1    | RW |
| 46417 | Phase 3 volt amps             | U32 | VA  | 1    | RW |
| 46419 | Phase 1 volt amps reactive    | S32 | Var | 1    | RW |
| 46421 | Phase 2 volt amps reactive    | S32 | Var | 1    | RW |
| 46423 | Phase 3 volt amps reactive    | S32 | Var | 1    | RW |
| 46425 | Phase 1 power factor          | S16 | -   | 0.01 | RW |
| 46426 | Phase 2 power factor          | S16 | -   | 0.01 | RW |
| 46427 | Phase 3 power factor          | S16 | -   | 0.01 | RW |
| 46428 | Phase 1 phase angle           | U16 | °   | 1    | RW |
| 46429 | Phase 2 phase angle           | U16 | °   | 1    | RW |
| 46430 | Phase 3 phase angle           | U16 | °   | 1    | RW |
| 46431 | Average line to neutral volts | U16 | V   | 0.1  | RW |
| 46432 | Average line current          | U16 | A   | 0.1  | RW |
| 46433 | Sum of line currents          | U16 | A   | 0.1  | RW |
| 46434 | Total system power            | S32 | w   | 1    | RW |
| 46436 | Total system volt amps        | U32 | VA  | 1    | RW |
| 46438 | Total system VAR              | S32 | Var | 1    | RW |
| 46440 | Total system power factor     | S16 | -   | 0.01 | RW |

|       |                               |     |      |      |    |
|-------|-------------------------------|-----|------|------|----|
| 46441 | Total systemphase angle       | U16 | °    | 1    | RW |
| 46442 | Frequency of supply voltages  | U16 | Hz   | 0.01 | RW |
| 46443 | Import Wh since last reset    | U32 | Wh   | 1    | RW |
| 46445 | Export Wh since last reset    | U32 | Wh   | 1    | RW |
| 46447 | Import Varh since last reset  | U32 | Varh | 1    | RW |
| 46449 | Export Varh since lastreset   | U32 | Varh | 1    | RW |
| 46451 | Phase 1 line to neutral volts | U16 | V    | 0.1  | RW |

(\*) ----- Supported on some models

31601~31681, 41102~41156 Special for storage inverter

### 3.4 Warning and Error Codes

| Warning Code | Description                          |
|--------------|--------------------------------------|
| 0            | No warning                           |
| 30           | Recover from warning                 |
| 150          | SPD Damaged                          |
| 156          | Internal fan warning                 |
| 157          | External fan warning                 |
| 163          | String current abnormal              |
| 165          | Ground connect warning               |
| 166          | CPU self-test ---- Register abnormal |
| 167          | CPU self-test ---- RAM abnormal      |
| 168          | CPU self-test ---- ROM abnormal      |
| 174          | Low Air Temprature                   |
| 175          | Battery Soc Low                      |
| 176          | Battery Fault Status                 |
| 177          | Battery Communication Disconnect     |
| 178          | EPS Output Over                      |
| 179          | Combox and Cloud Disconnect          |
| 180          | PV string inverse                    |



| Error Code | Description                               |
|------------|-------------------------------------------|
| 1          | Communication Fails between M-S           |
| 3          | Relay check Fail                          |
| 4          | DC Injection High                         |
| 5          | The result of Auto Test Function is fail  |
| 6          | DC bus is too high                        |
| 8          | AC HCT Failure                            |
| 9          | GFCI Device Failure                       |
| 10         | Device fault                              |
| 32         | ROCOF Fault                               |
| 33         | Fac Faulure :Fac Out of Range             |
| 34         | AC Voltage Out of Range                   |
| 35         | Utility Loss                              |
| 36         | GFCI Failure                              |
| 37         | PV Over Voltage                           |
| 38         | Isolation Fault                           |
| 40         | Over temperature in Inverter              |
| 41         | Consistent Fault :Vac differs for M-S     |
| 42         | Consistent Fault :Fac differs for M-S     |
| 43         | Consistent Fault :Groud I differs for M-S |
| 44         | Consistent Fault :DC inj. Differs for M-S |
| 45         | Consistent Fault :Fac,Vac differs for M-S |
| 46         | High DC bus                               |
| 47         | Consistent Fault                          |
| 48         | Average volt of ten minutes Fault         |
| 49         | PV1 lightning arrester fault              |
| 50         | PV2 lightning arrester fault              |
| 51         | Fuse failure                              |
| 52         | Neutral line loss fault                   |

|    |                                                                                                                                                                    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 53 | Insulation impedance test: before enabling the constant current source, the sampling value of insulation impedance measurement voltage is greater than 300mV       |
| 54 | Insulation impedance detection: after enabling constant current source, the sampling value of insulation impedance measurement voltage is out of range (1.37V±20%) |
| 55 | Insulation impedance detection: N-PE relay switches, and the instantaneous value of insulation impedance measurement voltage is less than 40mV                     |
| 56 | GFCI protect fault:30mA level                                                                                                                                      |
| 57 | GFCI protect fault:60mA level                                                                                                                                      |
| 58 | GFCI protect fault:150mA level                                                                                                                                     |
| 59 | PV1 string current is abnormal                                                                                                                                     |
| 60 | PV2 string current is abnormal                                                                                                                                     |
| 61 | DRMS Communication Fails(S9 Open)                                                                                                                                  |
| 62 | DRMS order disconnection device(S0 Close)                                                                                                                          |
| 63 | L-PE short-circuit protection error                                                                                                                                |
| 64 | PV input mode error                                                                                                                                                |
| 65 | PE connection Fault                                                                                                                                                |
| 66 | PV1 reverse connection fault                                                                                                                                       |
| 67 | PV2 reverse connection fault                                                                                                                                       |
| 68 | PV3 reverse connection fault                                                                                                                                       |
| 69 | External input failure                                                                                                                                             |
| 70 | AFCI self-test failure (including self-test circuit and CAN circuit failure)                                                                                       |
| 71 | AFCI failure (PV1-10)                                                                                                                                              |
| 72 | Parallel 485 communication fault                                                                                                                                   |
| 73 | Parallel CAN communication fault                                                                                                                                   |

### 3.5 Grid Codes

| Grid Code | Description             |
|-----------|-------------------------|
| 8         | GR PPC                  |
| 35        | NB/T32004:2018          |
| 47        | AU AS 4777.2 : 2015     |
| 48        | NZ AS 4777.2 : 2015     |
| 49        | ENGG-50Hz               |
| 50        | ENGG-60Hz               |
| 51        | TOR Erzeuger Typ A V1.1 |
| 59        | CNS15382:2018           |
| 64        | EN 50549-1              |
| 65        | NL EN50549-1:2019       |
| 66        | BR NBR 16149:2013       |
| 67        | VDE0126-1-1/A1/VFR      |
| 68        | IEC 61727 50Hz          |
| 69        | C10/11:2019             |
| 70        | VDE-AR-N4105:2018       |
| 71        | IEC 61727 60Hz          |
| 72        | G98/1                   |
| 73        | G99/1                   |
| 74        | AU AS/NZS4777.2:2020 A  |
| 75        | AU AS/NZS4777.2:2020 B  |
| 76        | AU AS/NZS4777.2:2020 C  |
| 77        | NZ AS/NZS4777.2:2020    |
| 78        | IL SI4777.3             |
| 79        | KR KS C 8565:2020       |
| 80        | ES UNE206007-1          |
| 81        | CY EN50549-1            |
| 82        | CS PPDS A1              |
| 83        | PL EN50549-1            |

|     |                           |
|-----|---------------------------|
| 84  | CEI 0-21:2019             |
| 85  | DK EN50549-1              |
| 86  | CH NA/EEA-NE7             |
| 87  | SE EIFS:2018              |
| 88  | FI EN50549-1              |
| 89  | RO Order208               |
| 90  | SI EN50549-1              |
| 91  | LV EN50549-1              |
| 92  | VDE0126/VFR2019 IS (50Hz) |
| 93  | VDE0126/VFR2019 IS (60Hz) |
| 94  | ZA NRS 097-2-1:2017       |
| 95  | BR PORTARIA No.140        |
| 96  | NTS 631 Type A            |
| 97  | NTS 631 Type B            |
| 98  | NO EN50549-1              |
| 99  | VDE-AR-N 4110             |
| 100 | EN 50549-2                |
| 101 | DEWA:2016                 |
| 102 | DK1 EN50549-1             |
| 103 | ZA RPPs                   |

### 3.6 Frame format

MODBUS protocol format: RTU format. Each communication data unit is composed of 1 bit starting bit, 8 bit data bit and 1 bit stopping bit, no parity .

MODBUS function codes:

- Read Holding Register (0x03)
- Read Input Register (0x04)
- Write Holding Single Register (0x06)
- Write Holding Multiple Registers (0x10)
- Write Holding Multiple Registers (0x10) for broadcast

#### 3.6.1 Read Holding Register (Function Code: 0x03)

**Request:**

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code        | 1 Byte |
| Register Address(Hi) | 1 Byte |
| Register Address(Lo) | 1 Byte |
| Register length      | 2 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

**Response:**

|               |            |
|---------------|------------|
| Device ID     | 1 Byte     |
| Function code | 1 Byte     |
| Byte count    | 1 Byte     |
| Data          | N × 1 Byte |
| CRC(Lo)       | 1 Byte     |
| CRC(Hi)       | 1 Byte     |

**Error:**

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code + 0x80 | 1 Byte |
| Exception code       | 1 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

3.6.2 Read Input Register (Function Code: 0x04)

Request:

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code        | 1 Byte |
| Register Address(Hi) | 1 Byte |
| Register Address(Lo) | 1 Byte |
| Register length      | 2 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

Response:

|               |            |
|---------------|------------|
| Device ID     | 1 Byte     |
| Function code | 1 Byte     |
| Byte count    | 1 Byte     |
| Data          | N × 1 Byte |
| CRC(Lo)       | 1 Byte     |
| CRC(Hi)       | 1 Byte     |

Error:

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code + 0x80 | 1 Byte |
| Exception code       | 1 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

### 3.6.3 Write Single Holding Register (Function Code: 0x06)

**Request:**

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code        | 1 Byte |
| Register Address(Hi) | 1 Byte |
| Register Address(Lo) | 1 Byte |
| Data                 | 2 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

**Response:**

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code        | 1 Byte |
| Register Address(Hi) | 1 Byte |
| Register Address(Lo) | 1 Byte |
| Data                 | 2 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

**Error:**

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code + 0x80 | 1 Byte |
| Exception code       | 1 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

3.6.4 Write Multiple Holding Registers (Function Code: 0x10)

Request:

|                      |           |
|----------------------|-----------|
| Device ID            | 1 Byte    |
| Function code        | 1 Byte    |
| Register Address(Hi) | 1 Byte    |
| Register Address(Lo) | 1 Byte    |
| Register length      | 2 Byte    |
| Data length          | 1 Byte    |
| Data                 | N × 1Byte |
| CRC(Lo)              | 1 Byte    |
| CRC(Hi)              | 1 Byte    |

Response:

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code        | 1 Byte |
| Register Address(Hi) | 1 Byte |
| Register Address(Lo) | 1 Byte |
| Register length      | 2 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |

Error:

|                      |        |
|----------------------|--------|
| Device ID            | 1 Byte |
| Function code + 0x80 | 1 Byte |
| Exception code       | 1 Byte |
| CRC(Lo)              | 1 Byte |
| CRC(Hi)              | 1 Byte |



**3.6.5 Write Multiple Holding Registers (Function Code: 0x10) for broadcast**

**Request:**

|                      |           |
|----------------------|-----------|
| Device ID            | 1 Byte    |
| Function code        | 1 Byte    |
| Register Address(Hi) | 1 Byte    |
| Register Address(Lo) | 1 Byte    |
| Register length      | 2 Byte    |
| Data length          | 1 Byte    |
| Data                 | N × 1Byte |
| CRC(Lo)              | 1 Byte    |
| CRC(Hi)              | 1 Byte    |

**Response: none**

**3.6.6 Exception Codes**

- 0x01 Illegal function
- 0x02 Illegal address
- 0x03 Illegal data
- 0x04 Slave device failure

## 4 Contact

If you experience any technical problems with our products, please contact the AISWEI Service Hotline to provide you with the necessary assistance:

**AISWEI Technology Co., Ltd.**

Room 905B, 757 Mengzi Road, Huangpu District  
200023 Shanghai(P.R. China)